

JKBOSE 2026

CLASS 11TH

Bhaichara on
TOP.

CHEMICAL BONDING AND MOLECULAR STRUCTURE

LECTURE-1

CHEMISTRY

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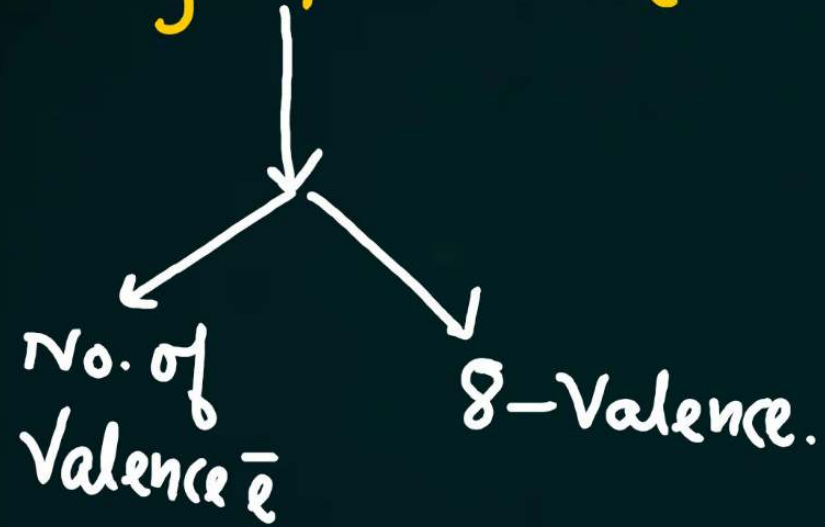
Topics To Covered

- 1** Kossel and Lewis Approach to Chemical Bonding OR Why Do Atoms Combine?
- 2** Lewis Dot Symbols
- 3** Ionic Bond

REVISION.

Valency

S & P } Representative



Diagonal Relationship



Oxides

ACIDIC

BASIC

AMPHOTERIC

NEUTRAL.

COMPARE the acid



KOSSEL AND LEWIS APPROACH TO CHEMICAL BONDING OR Why do atoms combine.

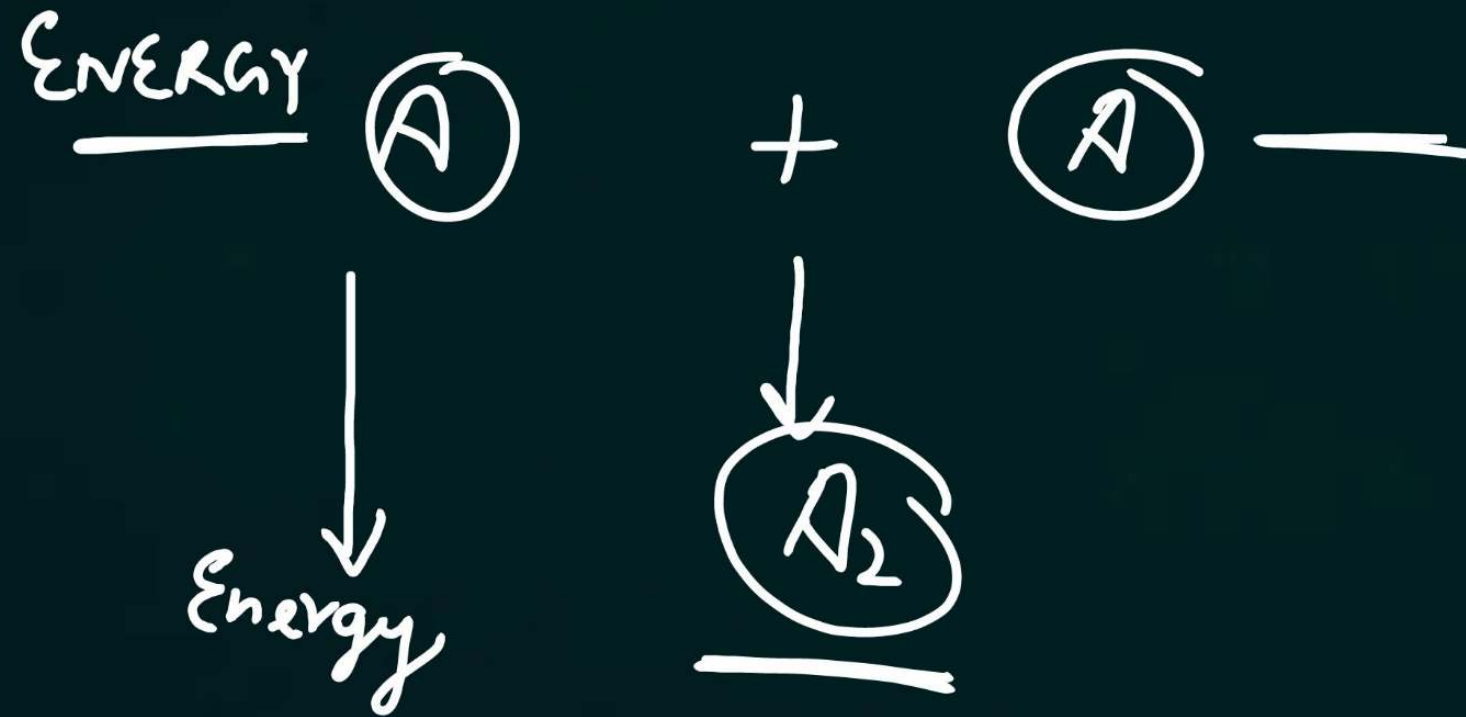
According to Kossel and Lewis atom consists of kernels (nucleus + inner e^-) and Valence electron. Atoms combine because of following reason.

a) To acquire noble gas configuration $\Rightarrow$
Noble gas have 8 e^- in their valence shell (except He) and they do not react. i.e. they are regarded as stable. Atoms combine to gain stability or to achieve noble gas configuration (i.e. 8 e^- in their valence shell).

"The principle of acquiring 8 e^- in their valence shell to gain stability is called octet Rule".

ii) To acquire minimum energy $\Rightarrow$

According to them when atoms combine to form molecule. their energy decreases. hence stability increases.



UNIT - IV: CHEMICAL BONDING AND MOLECULAR STRUCTURE**7 MARKS**

- Valence electrons, Ionic bond, covalent bond, bond parameters, Lewis structure, polar character of covalent bond, valence bond theory, resonance, geometry of covalent molecules. VSEPR theory, concept of hybridization involving s , p and d -orbitals and shapes of some simple molecules, molecular orbital theory of homonuclear molecules [Qualitative idea only], hydrogen bond.

Lewis Dot Symbols

As we know only valence e^- participate in bonding. G. N. Lewis gives a method of representing an element with their symbols & valence e^- as dots. are called Lewis dot symbols.

Element (Z)	Lewis dot symbol
Hydrogen(1)	$\cdot\text{H}\cdot$
Helium (2)	$\cdot\text{He}\cdot$
Lithium (3)	$\cdot\text{Li}\cdot$
Beryllium(4)	$\cdot\text{Be}\cdot$

Boron(5)	$\cdot\text{B}\cdot$
Carbon(6)	$\cdot\text{C}\cdot$
Nitrogen(7)	$\cdot\text{N}\cdot$
Oxygen(8)	$\cdot\text{O}\cdot$
Fluorine(9)	$\cdot\text{F}\cdot$
Neon (10)	$\cdot\text{Ne}\cdot$
Sodium (11)	$\cdot\text{Na}\cdot$
Magnesium (12).	$\cdot\text{Mg}\cdot$
Aluminium(13).	$\cdot\text{Al}\cdot$

HOMEWORK ✓.

Lewis dot symbol.



Silicon (14)

Phosphorus (15)

Sulphur (16)

Chlorine (17)



Argon. (18)

Potassium (19)

Calcium (20)





(ns^1) 1e	(ns^2)	<u>ROUGH</u> $\rightarrow$ $(ns^2 np^1)$		$(ns^2 np^2)$	$(ns^2 np^3)$	$(ns^2 np^4)$	$(ns^2 np^5)$	18 He (ns^2)
Group 1	2	13	14	15	16	17		
H								Ne
Li	Be	B	C	N	O	F		Ar
Na	Mg	Al	Si	P	S	Cl		Kr
K	Ca	Ga	Ge	As	Se	Br		Xe
Rb	Sr	In	Sn	Sb	Te	I		Rn
Cs	Ba	Tl	Pb	Bi	Po	At		
Fr	Ra	Nh	Fl	Mc	Lv	Ts		Og

IONIC BOND

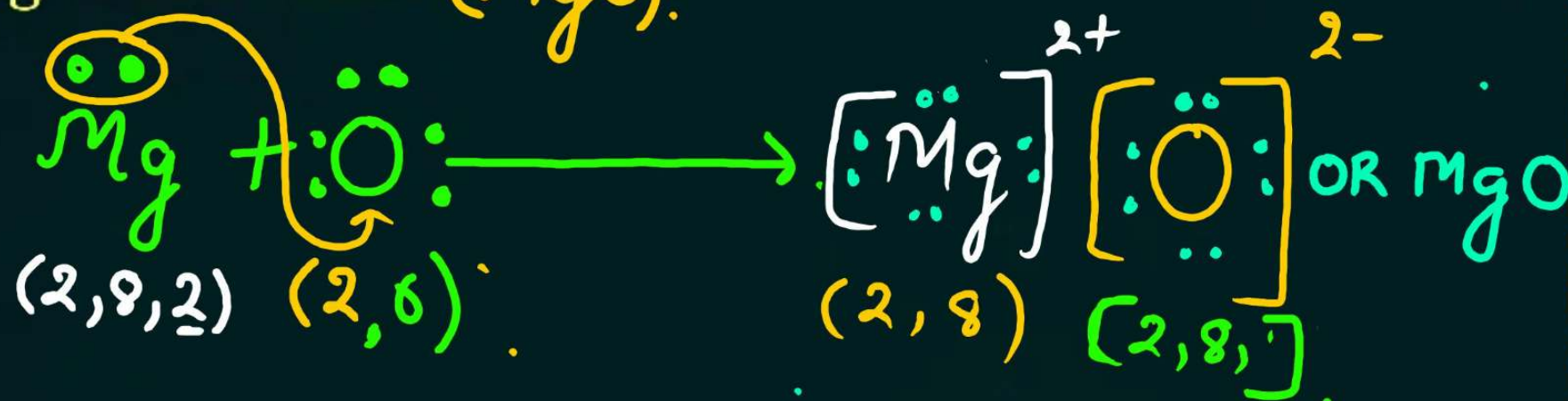


- When a bond is formed by the transference of one or more valence electrons of one atom to the other atom so as to complete their outermost octets and acquire stable nearest noble gas configurations, the bond formed is called electrovalent bond or ionic bond.
- In the formation of electrovalent bond, the number of electrons lost or gained by an atom is called its electrovalence or electro valency.
- The compounds containing ionic or electrovalent bonds are called ionic or electrovalent compounds.

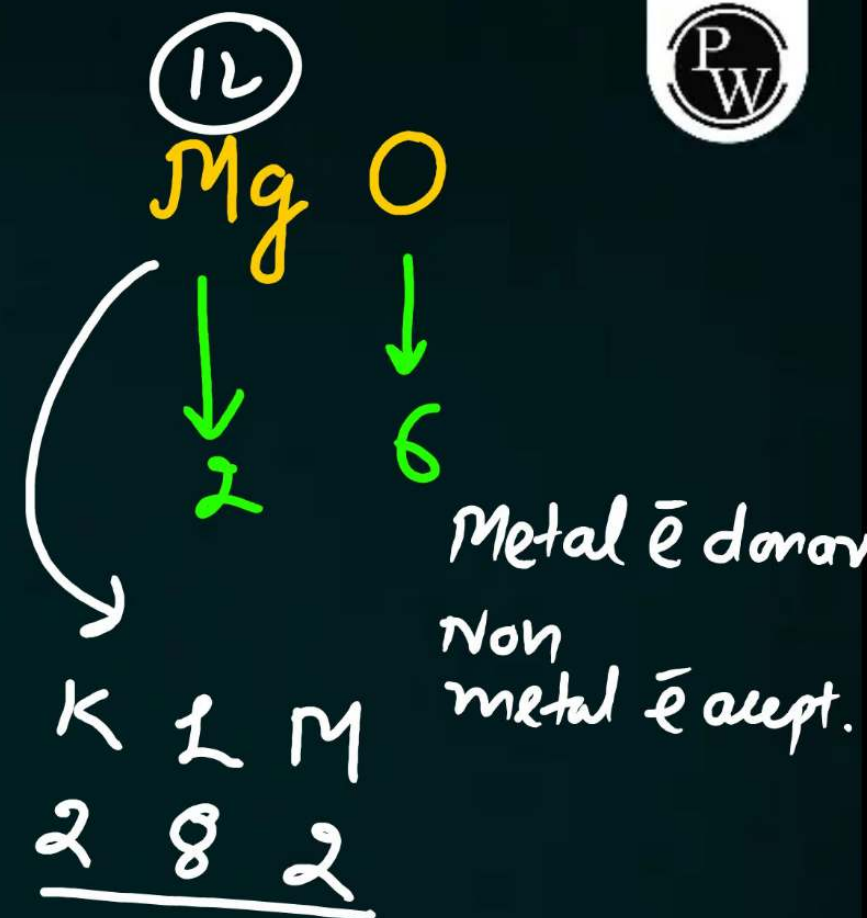
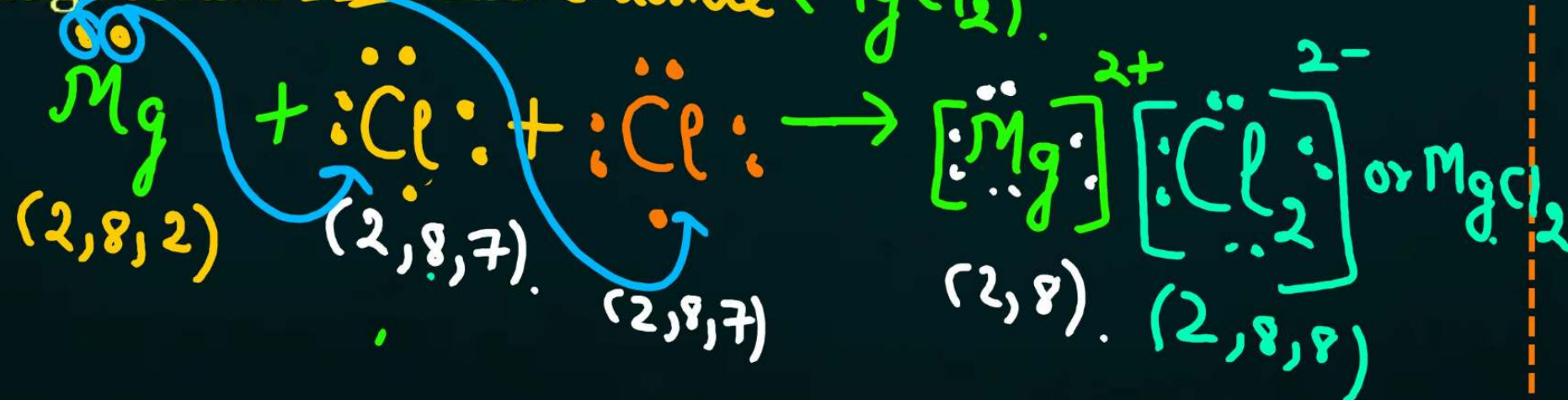
IONIC BOND

Some examples of formation of ionic bond are:

(i) Magnesium oxide: (MgO).



(ii) Magnesium ~~bromide~~ Chloride ($MgCl_2$).



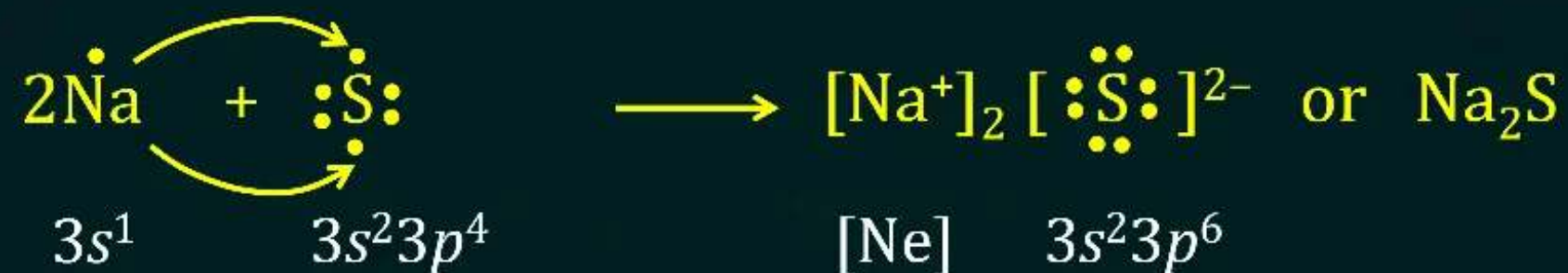
$$\begin{aligned} \text{No. of } \bar{e} \text{ in a shell} &= 2n^2 \\ &= 2 \times 1^2 \\ &= 2 \\ \text{2nd shell} &= 2 \times 2^2 \\ &= 8 \end{aligned}$$

IONIC BOND

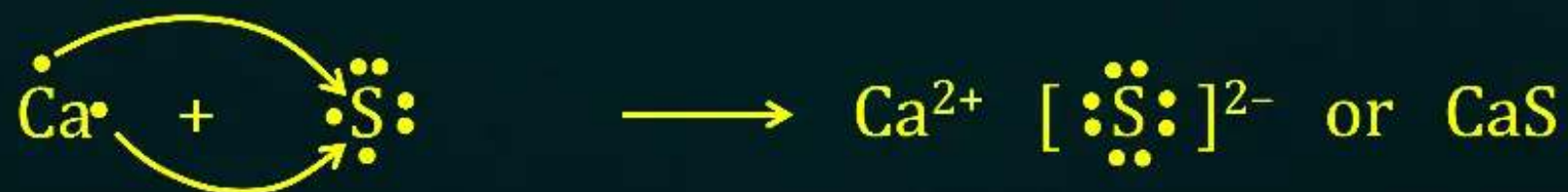


Some examples of formation of ionic bond are:

(i) Sodium sulphide:



(ii) Calcium sulphide:



QUESTION

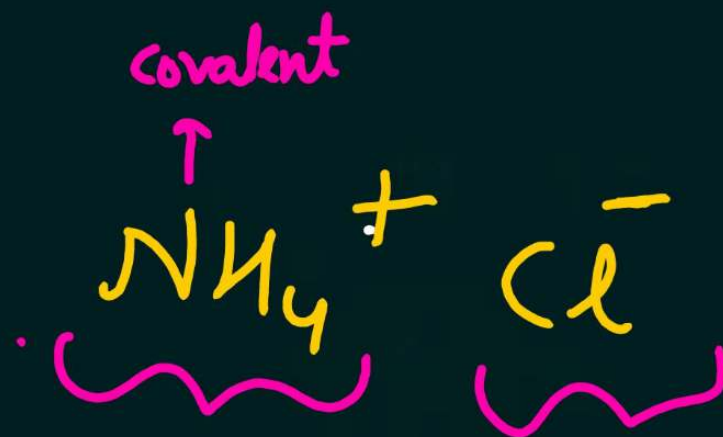
Which of the following contains both covalent and ionic bonds?

1 CCl_4 (covalent)

2 CaCl_2 (Ionic)

3 NH_4Cl

NER T



4 H_2O (covalent)

Electrovalency of Cl in $MgCl_2$

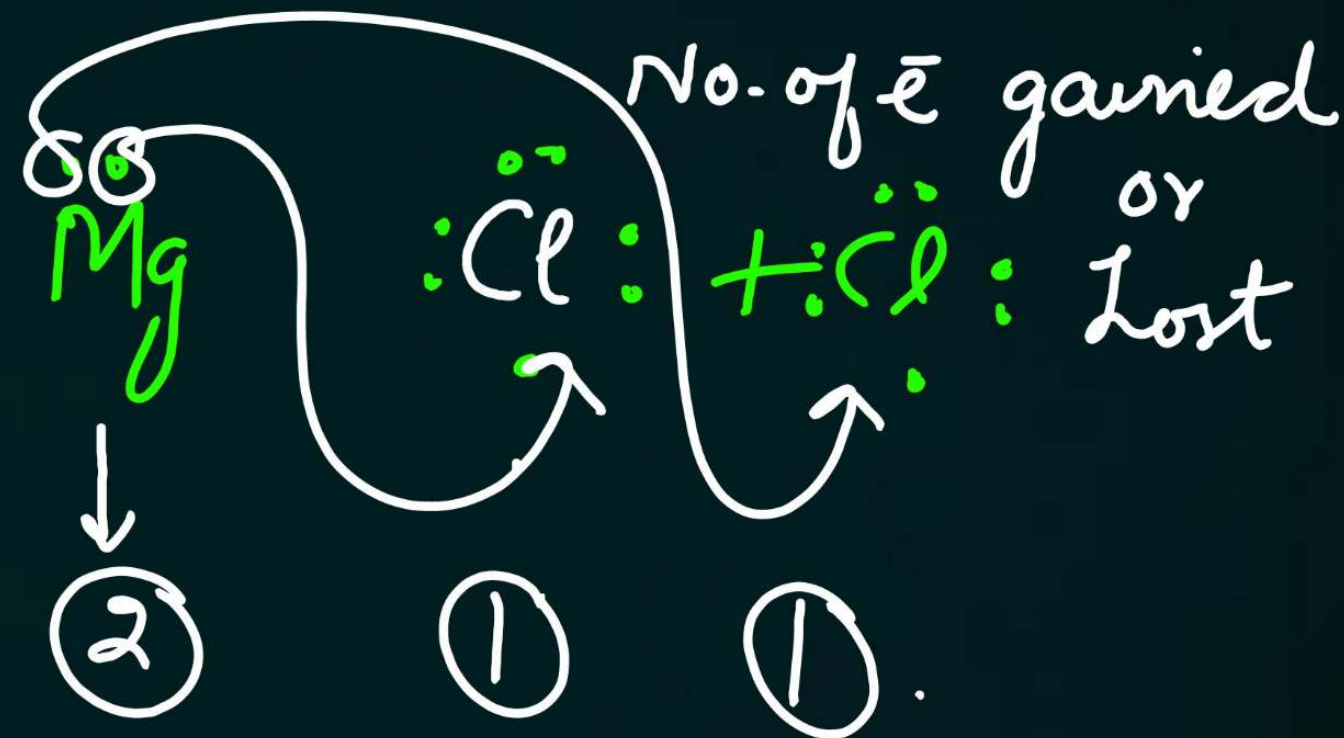
a) 1

b) 2

c) 3

d) 4

Electro
Valency

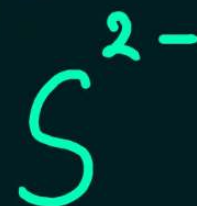


QUESTION



Write Lewis symbols for the following atoms and ions:

S and S^{2-} ; Al and Al^{3+} ; H and H^{-}

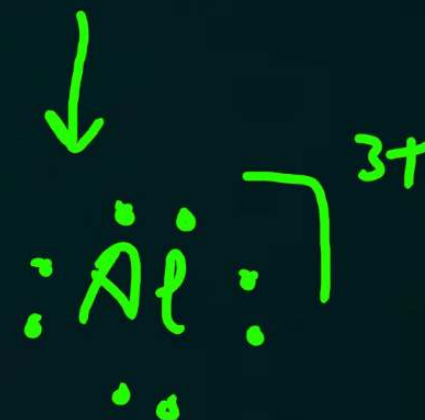
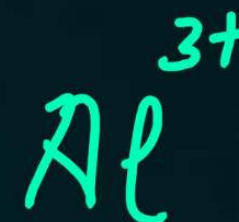


Se
Te
Po



(2, 8, 3).

OR



$$\text{H}$$


$$\dot{\text{H}}$$

$$\text{H}^-$$


$$\ddot{\text{H}}$$

QUESTION

Lewis has pictured the atom in terms of kernel and the outermost shell that could accommodate a maximum of

1 6 electrons

2 10 electrons

3 8 electrons

4 2 electrons

QUESTION

Which of the following elements would most likely form an ionic bond with chlorine (Cl)?

$\bar{e}$ donor $\bar{e}$ acceptor.
Metal + Non-metal

1 Oxygen (O)

2 Nitrogen (N)

3 Potassium (K)

4 Carbon (C)

Cl



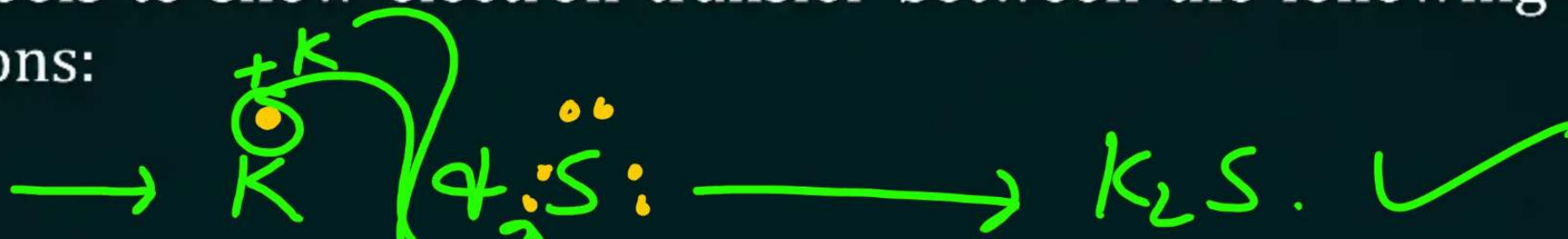
Metal + Non-metal

QUESTION



Use Lewis symbols to show electron transfer between the following atoms to form cations and anions:

(i) K and S



(ii) Ca and O



(iii) Al and N



REVISION KAR K SOO JANA

Thank
You

